Poltergeist

FM bass synth.



User manual · version 3.1.0 · VST3 · AU · CLAP · Standalone
GhostBlock

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Poltergeist is an FM bass and spirit synth. Six "ghosts" (FM operators) plus a NOISE ghost and a sample slot haunt each other to make growls, 808s, wobbles, reeses, bells and leads.

Version 3.1.0. Formats: VST3, AU, CLAP, Standalone. It is a stereo instrument (it takes MIDI in, audio out).

Quick start



Poltergeist

Make your first growl in 2 minutes

1. Load Poltergeist as an instrument on a track and play some low notes (the on-screen keyboard at the bottom works too).
2. Click the preset name at the top (it shows the current sound). Open a folder such as **WOBBLE** or **GROWL / SCREECH** and pick a sound. Use the **<** and **>** buttons to step through the list.
3. Or type words in the box at the top right, for example `dark growl`, and press **SUMMON**. Press **SUMMON** again for another take.
4. Turn the big **HAUNT** knob (bottom row). It sets how hard the ghosts hit each other, so the growl gets angrier.
5. Open the **WOBBLE** tab and drag the points on the drawn shape. The wobble follows your tempo.

6. Open the **BASS** tab. Turn on **SUB** to add a clean sine under the growl, and set **LOW MONO** to about 120 Hz to keep the low end mono.
7. Like it? Press **SNAPSHOT** (next to the **< >** buttons) to keep the sound, or **SAVE** to name it.
8. Lower **MASTER** (on the GHOSTS tab, VOICE card) if it clips. The default is -6 dB.

> Tip: **RANDOM** (top right) summons a surprise sound. Great for ideas.

The screen

The window is 1040 x 744 and you can resize it (from 75% to 200%, the shape stays the same). Everything works with the mouse.

- **Knobs.** Drag up or down. Hover to see a tooltip. Double-click to go back to the default. Right-click to type a value (for example `20 Hz` or `50%`).
- **Bipolar sliders** (in the MOD MATRIX): click to jump to a spot, **Shift** = fine, double-click = 0, right-click = type.

The header (always visible)

From left to right:

Control	What it does
POLTERGEIST logo and version	Shows the plugin version.
< / **>**	Previous / next preset. Shift-click jumps to the previous / next folder. When a snapshot is loaded, they step through your snapshots instead.
Preset name	Click for the preset menu (see Presets below).
Star	Mark the current preset as a favourite. Favourites show at the top of the preset list.
SAVE	Save this sound under a name (saved in Documents/Ghosty/Poltergeist).
LOAD	Load a sound you saved with SAVE.
SNAPSHOT and **v**	**SNAPSHOT** saves the sound right now into the SNAPSHOTS bank (Shift+Alt+S). **v** lists your snapshots: load, star, rename, delete, show the folder.
Text box + **SUMMON**	Type what you want (`dark trap bell`, `wobble growl`, `808 slide`). SUMMON makes a sound from your words. With an empty box it asks for words.
RANDOM	Summons a random sound and clears the text box.
BYPASS	Mutes the whole plugin. Click again to turn it back on.

Under that there is a row of **style chips**: **808, Bell, Growl, Wobble, Reese, Pluck, Keys, Pad, Choir, Lead, Brass, Riser, Arp**. Click one to summon that type. Next to them are modifier chips you can switch on: **dark, bright, haunted, dirty, wide, glitch, lofi, metallic**. Turn some on, then click a style chip to add those words.

Under the chips is the tab row: **GHOSTS, MOVE, FX, PLAY, BASS, WOBBLE, MORPH**. At the right end of the tab row are **UNLOCK** (demo only) and the look button (it names the next look). A status line between the tabs and those buttons tells you what just happened.

At the bottom of every tab:

- The six big **macro knobs**: **HAUNT, SPECTRE, POSSESS, DRIFT, VEIL, CURSE**.
- The **keyboard** (C1 to C6, with C3 as middle C). Click keys to play.

The macro knobs

All run from 0 to 100%.

Macro	What it does	Default
HAUNT	FM growl: how hard the ghosts hit each other.	0%
SPECTRE	Unison: more ghosts, wider and thicker.	0%
POSSESS	Drive: warm grit up to full possession.	0%
DRIFT	Movement: wobble, wow and vibrato.	0%
VEIL	Space: reverb and echo.	15%
CURSE	Glitch: crush, decimate and stutter.	0%

The mod wheel (CC1) adds to HAUNT and also opens the vowel of the TALK section.

Tab: GHOSTS

This is the FM synth. The left side is the **ghost field**. The right side is the panel for the ghost you selected. Under them are the MAIN FILTER, PITCH DROP, DRIFT (LFO) and VOICE cards.

The ghost field

You see ghosts 1 to 6, a **NOISE** ghost, and an **ADD SAMPLE** ghost (bottom right). A ghost labelled "you hear it" is sent to the output. A ghost labelled "haunts others" is only used to modulate other ghosts.

- **Click a ghost** to select it. Its controls show in the panel on the right.
- **Drag one ghost onto another** to make it HAUNT it (FM). A link appears with a number in a small circle (the depth, 0 to 100).
- **Drag the link's circle up or down** to change how hard.
- **Double-click the circle** to break the link.
- A ring over a ghost shows it has FEEDBACK.
- Ghost 1 starts as the one you hear; ghost 2 haunts ghost 1 at 35%.

- You can link any ghost (1 to 6, NOISE, or the SAMPLE) onto ghosts 1 to 6.

The ghost panel (GHOST 1 to 6)

Control	What it does	Range / default
Waveform menu	The ghost's waveform. **Classic Morph** is the original (WAVE morphs sine, triangle, saw, square). The other 31 are band-limited tables and WAVE bends them.	Classic Morph
Envelope picture	Shows the ghost's envelope. Click it to switch between **AMP** (the ghost's own) and **FILTER** (the filter envelope). A dot traces the notes you play.	
RATIO	The ghost's pitch relative to the note. Snaps to musical ratios; hold **Shift** to move freely.	0.25 to 16. Defaults: 1, 2, 1, 3, 0.5, 4 for ghosts 1 to 6
FINE	Fine tune.	-50 to +50 cents, 0
OUT	How loud you hear this ghost. At 0 it is only heard through the ghosts it haunts.	0 to 100%. Ghost 1: 80%, others 0
WAVE / BEND	Waveform morph (Classic Morph) or phase bend (the tables).	0 to 3, default 0
FEEDBACK	The ghost haunts itself.	0 to 100%, 0
VELOCITY	How much velocity changes this ghost.	0 to 100%, 50%
DELAY	Time before the envelope starts.	0 to 4 s, 0
ATTACK	Fade-in.	1 ms to 10 s, 3 ms
HOLD	How long it stays at full level after the attack.	0 to 4 s, 0
DECAY	Fall to the sustain level.	5 ms to 10 s, 0.6 s
SUSTAIN	Level while the key is held.	0 to 100%. Ghost 1: 80%, others 40%
RELEASE	Fade-out after you let go.	5 ms to 10 s, 0.35 s
LOOP ENV	Loops the envelope (attack, hold, decay) while the key is held: growls that talk by themselves.	off
TO FILTER 2	With the filters on **SPLIT** routing, this ghost goes through FILTER 2 instead of FILTER 1.	off
Scope	A small live picture of the output wave.	

The NOISE ghost panel

Select the **NOISE** ghost. Drag it onto a ghost to add noise FM: grit, breath and crackle inside the growl.

Control	What it does	Range / default
Type menu	**White, Pink, Crackle**.	White
OUT	How loud you hear the noise.	0 to 100%, 0
CUTOFF	Filter for the noise.	20 Hz to 20 kHz, 20 kHz
RESO	Filter resonance.	0 to 100%, 10%
Filter menu	**Low, Band, High**.	Low
ATTACK / DECAY / SUSTAIN / RELEASE	The noise envelope.	1 ms, 0.3 s, 100%, 0.2 s
TO FILTER 2	Send the noise through FILTER 2 (SPLIT routing).	off

The ADD SAMPLE ghost

This is the one sample feature. A sample becomes an FM source. It can haunt a ghost, be heard itself, or be used as a ghost's waveform.

1. Click the **ADD SAMPLE** ghost, or press **ADD SAMPLE** in its panel, or just drop a WAV, AIFF, FLAC or MP3 file anywhere on Poltergeist. (The file chooser lists wav, aif, aiff, flac and mp3. Dropping also accepts ogg.)
2. Poltergeist looks for the pitch of the sample and sets **ROOT** to it. If it cannot find a clear pitch, ROOT is C3.
3. If the sample is still silent (LEVEL and HAUNT at 0), it is set to LEVEL 50% and HAUNT 30% so you hear something at once. It loops by default.

Control	What it does	Range / default
ADD SAMPLE / **REPLACE**	Pick a file. The button says REPLACE when a sample is loaded.	
CLEAR	Remove the sample (the ghosts keep playing).	
ROOT	The note the sample is in. The key you play then sounds that note. Pick **AUTO** to find it again.	C1 to C5
Waveform	Drag **START** and **END**. Right-click = START here. Right-drag = LOOP (two fingers on a trackpad). Click the wave to hear it.	

Control	What it does	Range / default
LOOP ON / LOOP OFF (button on the waveform)	ON: it repeats while you hold a key. OFF: one-shot.	ON after adding
LEVEL	How loud you hear the sample itself. 0 = you only hear what it does to the ghost it haunts.	0 to 100%, 0
HAUNT	How hard the sample modulates its ghost (FM). A vocal or a growl here makes the FM voice talk.	0 to 100%, 0
ATTACK	Fade-in for each note.	2 ms
RELEASE	Fade-out after you let go.	0.2 s
HAUNTS	Which ghost (1 to 6) the sample modulates. You can also drag the SAMPLE ghost onto a ghost. That ghost must be heard (OUT) or haunt one that is.	Ghost 1
AS WAVE OF	That ghost plays one cycle of the sample as its waveform. START picks which moment. Off = its own waveform. A dashed **WAVE** line shows the link on the field.	Off

The sample's waveform has no export grip. Poltergeist reads files in only; it does not write them out.

> Old projects and sounds that used the removed sample pages keep working. If a sound carries an old sample mode, mute or reverse setting, a note under the waveform says so. It still plays as saved.

Cards under the field

MAIN FILTER (the simple filter, always on)

Control	What it does	Range / default
Type menu	**Low, Band, High**.	Low
CUTOFF	Filter frequency.	20 Hz to 20 kHz, 20 kHz (open)
RESO	Resonance.	0 to 100%, 10%
ENV	How much the filter envelope opens the filter (negative closes).	-100% to +100%, 0
KEY	How much the cutoff follows the note you play.	0 to 100%, 30%

PITCH DROP (a quick pitch fall at the start of each note: the 808 "boom")

Control	What it does	Range / default
AMOUNT	How far the pitch starts above (or below) the note.	-48 to +48 semitones, 0
DECAY	How fast it falls.	5 ms to 2 s, 60 ms

DRIFT (LFO) (a slow movement source used by the macros, TALK and the MOD MATRIX)

Control	What it does	Range / default
RATE	Speed.	0.02 to 30 Hz, 2 Hz
SYNC	Lock the speed to the tempo.	off
Division menu	Used when SYNC is on: 1/1, 1/2, 1/4, 1/4T, 1/8, 1/8T, 1/8D, 1/16, 1/16T, 1/32.	1/8
Shape menu	Sine, Triangle, Saw, Square, Random.	Sine

VOICE

Control	What it does	Range / default
MONO	One note at a time.	off
RETRIG	In mono: every new note restarts the envelopes, the sub and a retriggered wobble. Off = overlapping notes glide legato.	off
GLIDE	Slide time between notes.	0 to 1 s, 0
OCTAVE	Shift the whole synth.	-3 to +3, 0
VOICES	How many notes at once.	1 to 16, 8
MASTER	Output level.	-24 to +6 dB, -6 dB

The pitch wheel bends notes. The bend range is a saved setting (**Bend Range**, 0 to 24 semitones, default 2) that shows up as a parameter in your DAW's automation list, but there is no knob for it on screen. The sustain pedal holds notes (on sounds that use the newer engine revision; older presets keep the original engine, which ignores the pedal).

Tab: MOVE

Three cards: RHYTHM, TALK (VOWEL FILTER), BUBBLES.

RHYTHM. Draw 16 steps with your mouse (click or drag across the bars). The playing step lights up.

Control	What it does	Range / default
RHYTHM ON	Turns the rhythm on.	off
Rate menu	Step length (same list as the DRIFT divisions).	1/16
Target menu	What the rhythm moves: **Gate** (stutter), **Filter**, **Haunt** (FM growl), **Talk** (vowels) or **Pitch** (jumps).	Gate
AMOUNT	How strongly.	0 to 100%, 100%
SMOOTH	Smooths the jumps between steps.	0 to 100%, 10%
STEPS	How many steps are used.	2 to 16, 16

TALK (VOWEL FILTER). Makes the bass speak (a, e, i, o, u).

Control	What it does	Range / default
TALK	How much of the vowel filter you hear.	0 to 100%, 0
VOWEL	Which vowel.	0 to 100%, 0
MOVE	Lets the DRIFT LFO (or the rhythm when its target is Talk) work the mouth.	0 to 100%, 50%
RESO	Sharpness of the vowel.	0 to 100%, 60%

BUBBLES. A pitch-dropping blip on every step: bubbly, poppy sparkle.

Control	What it does	Range / default
BUBBLES	Amount.	0 to 100%, 0
POP	How far each blip falls.	0 to 48 semitones, 24
TONE	Brightness.	0 to 100%, 40%
Rate menu	Blip rate.	1/16

Tab: FX

Card	Control	What it does	Range / default
COMB	**COMB**	A tuned metal resonance that tears through growls.	0 to 100%, 0

Card	Control	What it does	Range / default
	FEEDBACK	How long it rings.	0 to 97%, 75%
	TUNE	Pitch offset of the comb.	-24 to +24 semitones, 0
	FOLLOW KEY	The comb follows the note you play.	on
SYNC	**SYNC**	Ghost 1 hard-sync screech. Turn it with HAUNT for laser leads.	0 to 100%, 0
SQUEEZE	**SQUEEZE**	Three-band squash (down above, up below): the loud, bright, in-your-face finish.	0 to 100%, 0

Drive, crush and space stay on the big macros (POSSESS, CURSE, VEIL).

Tab: PLAY

Play helpers that change the notes you play before the synth hears them.

Card	Control	What it does	Default
SCALE LOCK	**SCALE LOCK**	Snaps every key into the key and scale, so nothing sounds wrong.	off
	Key menu	C to B.	C
	Scale menu	Major, Minor, Harmonic minor, Dorian, Phrygian, Pentatonic minor, Pentatonic major, Blues.	Minor
CHORD (ONE FINGER)	Chord menu	One finger plays a chord: **Off, Power, Triad, 7th, 9th, Sus2, Sus4, Octaves**. In key when SCALE LOCK is on.	Off
ARPEGGIATOR	**ARP**	Plays the notes you hold one at a time, in time with your song.	off
	Rate menu	Step length (1/1 to 1/32, with triplets and dotted 1/8).	1/16
	Pattern menu	**Up, Down, Up-Down, Random, As played**.	Up
	OCTAVES	Repeat over this many octaves.	1 to 3, 1
	GATE	Note length as a part of the step.	10% to 100%, 60%

Tab: BASS

Four cards: FILTERS, SUB + LOW END, DIST, MULTIBAND DRIVE.

The signal order is: ghosts, filters, POSSESS, DIST, MULTIBAND, comb, SQUEEZE, crush, space.

FILTERS. Two extra filters (separate from the MAIN FILTER on the GHOSTS tab). A small picture shows their combined response (**FILTERS OFF** when both are off).

Control	What it does	Range / default
FILTER 1 / **FILTER 2**	Switch each on.	off
Type menu (each)	Low 12, Low 24 Ladder, Band, High 12, High 24, Notch, Comb +, Comb -, Formant, Dirty.	Filter 1: Low 24 Ladder. Filter 2: Formant
Routing menu (top right of the card)	**Series** (1 into 2), **Parallel**, **Split** (ghosts marked TO FILTER 2 go through filter 2, the rest through filter 1).	Series
CUTOFF (each)	Frequency. With the **Formant** type this knob is called **VOWEL** (A, E, I, O, U). With a **Comb** type it is called **PITCH** (a note; set KEY to 100% to follow the note you play).	20 Hz to 20 kHz. Filter 1: 800 Hz, filter 2: 1 kHz
RESO	Resonance.	0 to 100%, 30%
DRIVE	Grit into the filter.	0 to 100%, 0
KEY	Follows the note.	0 to 100%, 0
ENV	Filter envelope amount.	-100% to +100%, 0

SUB + LOW END.

Control	What it does	Range / default
SUB	Adds a sub oscillator.	off
CLEAN SUB	The sub skips the filters and every effect.	on
SUB SOLO	Plays only the sub, to check it.	off
Wave menu	Sine, Triangle, Square, Rounded.	Sine
Octave menu	-1 oct or -2 oct.	-1 oct
FOLLOWS menu	Which envelope shapes the sub: **Ghost 1**, **Longest Ghost** (a long sub under a short pluck), or **Own**.	Ghost 1
LEVEL	Sub volume.	0 to 100%, 70%
LOW-PASS	Cuts the top of the sub.	40 Hz to 2 kHz, 400 Hz

Control	What it does	Range / default
LOW MONO	Everything below this frequency is mono. Under 20 Hz shows "Off".	0 to 300 Hz, Off
ATTACK / RELEASE	The sub's own envelope. Only active when FOLLOWS is **Own**.	2 ms / 0.12 s

DIST.

Control	What it does	Range / default
DIST	Turns the distortion on.	off
Type menu	Overdrive, Tube, Hard Clip, Fold, Bitcrush, Downsample.	Overdrive
DRIVE	Amount.	0 to 100%, 50%
TONE	Brightness.	0 to 100%, 100%
MIX	Dry / wet.	0 to 100%, 100%
QUALITY	How many times faster than your project the voices, FM and drive run while you play. Higher = cleaner highs on hard FM, sync and drive, more CPU.	2x, 4x, 8x. Default 2x
RENDER	The quality used when your DAW bounces offline.	Same as live, 4x, 8x. Default Same as live

MULTIBAND DRIVE. Each band gets its own drive. Leave LOW at 0 to keep the sub clean while the mids scream.

Control	What it does	Range / default
MULTIBAND	Turns it on.	off
LOW / MID / HIGH	Drive per band.	0 to 100%. 0, 50%, 40%
LOW X	Crossover between low and mid.	60 to 400 Hz, 150 Hz
HIGH X	Crossover between mid and high.	800 Hz to 6 kHz, 2.5 kHz
MIX	Dry / wet.	0 to 100%, 100%

Tab: WOBBLE

Three cards: WOBBLE (TEMPO-SYNCED, DRAWN), LFO 1 / LFO 2, MOD MATRIX.

WOBBLE. Draw a shape that repeats in time with your tempo. A label in the corner says how long one cycle is.

- Click an empty spot to add a point (up to 8).
- Drag a point to move it.
- Double-click a point to remove it (at least 2 stay).
- The first time you draw and nothing is using the wobble yet, Poltergeist points it at something you can hear: the filter cutoff (if a filter is on), or the volume.

Control	What it does	Range / default
Rate menu	Length of one cycle: 4/1, 2/1, 1/1, 1/2, 1/2T, 1/2D, 1/4, 1/4T, 1/4D, 1/8, 1/8T, 1/8D, 1/16, 1/16T, 1/16D, 1/32, 1/32T.	1/8
RETRIG	The wobble restarts on every note.	off
SHAPES	A menu of ready-made shapes.	
TARGET	What the wobble moves (the same list as the MOD MATRIX targets; **Volume** means the shape is the volume).	Off (set for you when you first draw)
DEPTH	How far it moves the target.	-100% to +100%, 0
SMOOTH	Rounds the corners of the shape.	0 to 100%, 100%

LFO 1 / LFO 2. Two more LFOs for the MOD MATRIX.

Control	What it does	Range / default
RATE	Speed.	0.02 to 30 Hz. LFO 1: 1 Hz, LFO 2: 0.25 Hz
SYNC	Lock to tempo.	off
Division menu	Same list as the wobble rate.	1/4
Shape menu	Sine, Triangle, Saw Up, Saw Down, Square, Random, Smooth Random.	Sine

MOD MATRIX. Eight slots numbered 1 to 8. Each slot has a **source** menu, a **target** menu and an amount slider.

- Sources: Off, Wobble, LFO 1, LFO 2, Drift LFO, Filter Env, Ghost 1 Env, Velocity, Mod Wheel, Key, Random.
- Targets: Off, Filter 1 Cutoff, Filter 2 Cutoff, Filters 1+2 Cutoff, Filter 1 Reso, Filter 2 Reso, Vowel, FM Amount, Ghost 1 Level to Ghost 6 Level, All Ghost Levels, Pitch, Noise Level, Sub Level, Dist Drive, Wave Bend, Main Filter Cutoff, Volume.

- Amount: -100% to +100% (default 0). Right = positive, left = negative.
- Cutoffs move up to 6 octaves. Pitch moves 24 semitones. Ghost Levels scale the ghost's OUT and its FM depth, so a ghost at OUT 0 is only heard through the ghosts it haunts.

Tab: MORPH

The MORPH board blends whole sounds. Each ghost on the board holds a complete sound. A glowing puck mixes them by distance.

- **Click the board** to add a ghost that holds the sound you hear now.
- Drag ghosts and the puck. Double-click a ghost to hear it alone.
- Right-click a ghost to set it from this sound, a snapshot or a preset.
- Up to 12 ghosts.

Control	What it does
MORPH ON	The board plays the blend. Off = the knobs' sound.
X / Y	The puck position (you can automate these in your DAW). Default 0.5 and 0.5.
ADD THIS SOUND	Adds a ghost holding the sound you hear now.
RANDOM BOARD	Scatters the ghosts.
ARRANGE IN CIRCLE	Spaces the ghosts evenly round the middle.
MIX menu	How the ghosts mix: **Smooth** (all, by distance), **Islands** (only near ones), **Nearest 3**.
ORBIT menu	The puck moves by itself: **Off, Circle, Figure 8, Random walk**.
SYNC	The orbit follows the tempo.
RATE	Orbit speed: cycles per beat (SYNC) or per second.
SIZE	Orbit size.
BLEND TO KNOBS	The blend at the puck becomes the knobs' sound (MORPH turns off). Then SAVE or SNAPSHOT it.
SAVE BOARD / LOAD BOARD	Save or load the board (ghosts and their sounds) under a name.
CLEAR	Removes every ghost.

Knob values glide between the ghosts. Switches and the sample follow the closest ghost.

Presets, favourites, snapshots

Click the preset name to open the menu:

- **SEARCH...** finds presets by words from the name, folder or style.
- **SNAPSHOTS** lists the sounds you took with SNAPSHOT.
- **FAVOURITES** lists the sounds you starred.
- **ORIGINAL** holds the first set of presets.
- **BASS: ...** folders: GROWL / SCREECH, RIDDIM, WOBBLE, REESE / NEURO, ROLLING / CLEAN, SUB, 808, MID-BASS PLUCK, LEADS.

Workflows

Trap 808 slide

1. Press the **808** style chip and play low notes.
2. On the GHOSTS tab, set **MONO** on, and raise **GLIDE** a little (50 to 120 ms) so notes slide.
3. In **PITCH DROP**, set **AMOUNT** to about +24 and **DECAY** to about 50 ms for the "boom".
4. On the BASS tab, turn **SUB** on, keep **CLEAN SUB** on.
5. Overlap two notes in your piano roll. With **RETRIG** off, overlapping notes glide legato (a slide).

Dubstep wobble that follows the tempo

1. Pick a preset from the **WOBBLE** folder.
2. On the WOBBLE tab, set the rate menu to 1/8 (or 1/4 for slower).
3. Draw a shape. Use **SHAPES** for ready-made ones.
4. Check **TARGET** and **DEPTH**. Push DEPTH up for a stronger wobble.
5. Turn on **RETRIG** if each note should start the wobble from the top.

Talking growl

1. Start with a **GROWL / SCREECH** preset.
2. Go to MOVE. Raise **TALK** and move **VOWEL**.
3. Raise **MOVE** so the DRIFT LFO moves the vowel on its own, or set the RHYTHM **target** to **Talk** and draw steps.
4. Or on the BASS tab set a filter type to **Formant** and move its **VOWEL** knob with the wobble (MOD MATRIX).

Reese with a clean sub

1. Pick a **REESE / NEURO** preset.
2. BASS tab: turn **SUB** on, **CLEAN SUB** on.
3. Set **LOW MONO** to about 120 Hz.
4. Turn **MULTIBAND** on and keep **LOW** at 0, so the sub stays clean while mids and highs get dirty.

Put a vocal into the FM

1. Drop a WAV of a voice on Poltergeist.
2. Select the **HAUNTS** target you want (for example Ghost 1). Raise **HAUNT** on the sample panel.
3. Make sure Ghost 1 is heard (OUT up) or haunts a ghost that is.
4. Play notes. The voice shapes the FM, so the bass "talks".
5. Raise **LEVEL** if you also want to hear the sample itself.
6. Use **AS WAVE OF** to turn one cycle of the sample into a ghost's waveform.

Blend two sounds with the MORPH board

1. Make a sound you like. On MORPH, press **ADD THIS SOUND**.
2. Load or build another sound. Press **ADD THIS SOUND** again (or right-click a ghost and set it from a preset).
3. Turn **MORPH ON** and drag the puck. Try **ORBIT: Circle** with **SYNC** for a moving blend.
4. Press **BLEND TO KNOBS** to keep the blend as a normal sound, then **SNAPSHOT** it.

Play a bass line in key

1. PLAY tab: turn **SCALE LOCK** on and set the key and scale.
2. Optional: pick a **CHORD**, or turn on **ARP** to get a pattern from held notes.
3. Play any keys. They all land in the scale.

FL Studio tips

- **Loading.** Add Poltergeist as an instrument (a "generator") from the plugin database. If you cannot find it, rescan plugins (see Troubleshooting).
- **MIDI.** Poltergeist is an instrument. It needs MIDI notes from a piano roll or a keyboard. It does not output MIDI. There is no MIDI or audio drag-out in Poltergeist. To get audio, bounce or render in FL Studio as usual.
- **Tempo sync.** The wobble, the RHYTHM, the ARP, the LFO divisions and the MORPH orbit follow your project tempo when they are in sync mode. When the host is not playing, the ARP uses its own clock.
- **Automation.** All the knobs, menus and switches show up as parameters. In FL Studio, right-click a knob and choose "Create automation clip". The **MORPH X / Y** puck position can be automated too.
- **Presets and sessions.** Your sound is saved with the project. **SAVE** writes a named sound file you can load in any project. **SNAPSHOT** keeps versions as you tweak. **SAVE BOARD** saves a MORPH board.
- **Latency.** The plugin reports its oversampling latency to the host. A higher **QUALITY** (4x, 8x) costs more CPU. Use **RENDER** to set a different quality for bounces.
- **Polyphony and CPU.** Lower **VOICES** and **QUALITY** if your CPU is busy.
- **Window size.** Drag the corner to resize the window.

Where files live

What	Mac	Windows
Saved sounds (`.ghostpatch`), `favourites.txt`	`~/Documents/Ghosty/Poltergeist`	`Documents\Ghosty\Poltergeist`
Saved boards (`.ghostboard`)	`~/Documents/Ghosty/Poltergeist/Boards`	`Documents\Ghosty\Poltergeist\Boards`
Snapshots (`.xml`)	`~/Library/Application Support/GhostBlock/Poltergeist/Snapshots`	`%APPDATA%\GhostBlock\Poltergeist\Snapshots`
Licence file	`~/Library/Application Support/GhostBlock`	`%APPDATA%\GhostBlock`

In the SNAPSHOT list (**v**), "Show the folder" opens the snapshots folder for you. I did not find a log file location in the code.

The looks

Poltergeist has three looks: **BLACK**, **PURPLE** and **LIGHT**. It does not have the fourth, glass-style GHOST look.

The look button is at the right end of the tab row (next to UNLOCK). Its text is the name of the next look. Tooltip: "Switch the look: BLACK -> PURPLE -> LIGHT (the button names the next look)". Click it to go to the next one. The look is saved with the project and is not part of any sound.

Demo vs licensed

In demo mode (no licence key entered):

- The sound drops out for 2 seconds every 20 seconds.
- A project you reopen starts at the default sound. Unlocking brings the saved sound back.
- Saves (your sounds, snapshots, boards, the project) leave out audio that the plugin made itself. Your own imported samples are still saved. A hint near **UNLOCK** tells you when something was left out.
- Poltergeist has no audio or MIDI drag-out, so there is no export limit to hit.

Everything else works in the demo.

To unlock: click the **UNLOCK** button in the top bar (right end of the tab row). A box asks for your licence key (the one from your receipt email). Paste it and press **Unlock**. **Buy** opens the store page. After unlocking, the button reads **LICENSED**. Click it to see which key is in use. The licence is re-checked quietly in the background about every 14 days.

Shortcuts

Keys	What it does
Shift+Alt+S (Option+Shift+S on a Mac)	Take a SNAPSHOT (when no text box has focus).
Ctrl/Cmd+Z	Undo the last snapshot delete (within 60 seconds).
Enter	In the SUMMON box: summon.
Shift-click **<** / **>**	Jump to the previous / next preset folder.
Shift while turning **RATIO**	Move freely instead of snapping to musical ratios.
Double-click a knob	Back to the default.
Right-click a knob	Type a value.
Shift while dragging a MOD MATRIX slider	Fine adjust.
Double-click a link circle	Break an FM link.
Double-click a wobble point	Remove it.
Mod wheel	Adds to HAUNT and to the vowel.
Pitch wheel	Bends the notes.
Sustain pedal	Holds notes (newer-engine sounds only).

Plain letter keys are not used on purpose, because hosts play the computer keyboard as piano keys.

Troubleshooting

1. **No sound.** Check that **BYPASS** is off. Check that a ghost is heard: **OUT** above 0 on a ghost (ghost 1 is 80% by default), or SUB on, or NOISE OUT up. Check **MASTER**. Check that notes reach the plugin: click a key on the on-screen keyboard to test the sound itself.
2. **Sound only through a haunted ghost.** A ghost with OUT at 0 is silent on its own. The ghost it haunts must be heard.
3. **No MIDI.** Make sure the track sends MIDI to Poltergeist. In FL Studio, route your controller to the channel that holds the plugin. Poltergeist is an instrument, not an effect.
4. **Demo dropout.** A 2-second silence every 20 seconds is the demo. Click **UNLOCK**.
5. **Sound resets when I reopen a project.** That is the demo behaviour. Unlock brings the saved sound back.
6. **Gatekeeper warning on a Mac.** Until the installer is notarized, right-click the installer or plugin and choose **Open**, then **Open** again.
7. **Plugin not showing in FL Studio.** Make sure you installed the format your host uses (VST3, AU, CLAP). In FL Studio open the plugin manager and run **Find installed plugins** (a rescan). Restart FL Studio if needed.
8. **Highs sound harsh or aliased on hard FM.** Raise **QUALITY** on the BASS tab (4x or 8x). It costs CPU.
9. **The sample does nothing.** Both **LEVEL** and **HAUNT** may be 0, or the **HAUNTS** ghost is not heard. Raise LEVEL to hear it, and make sure the haunted ghost is heard.
10. **A file does not load.** Use WAV, AIFF, FLAC or MP3. A file with no sound in it is refused, and the status line says so.

FAQ

Is there a drag-out for audio or MIDI? No. Poltergeist does not export files. Use your DAW to bounce audio.

Can I use my own samples? Yes: one sample, with ADD SAMPLE. It becomes an FM source. Sample editing, chopping and slicing are not in Poltergeist.

How many sounds does it come with? An ORIGINAL set plus the BASS folders. The preset menu lists them all, and SEARCH finds them by name or style.

What does SUMMON do? It matches your words to a style (for example "808", "growl", "bell") and builds a sound from a recipe. A new tap gives a new variation. Modifier words such as dark, bright, haunted, dirty, wide, glitch, lofi and metallic change the recipe.

How do I keep a sound I tweaked? Use **SNAPSHOT** for a quick keep, or **SAVE** to name it. Both can be loaded in any project.

Why does the bass change when I press a key higher on the keyboard? The filter **KEY** knobs and the comb can follow the note. Set KEY to 0 for a fixed filter.

Can I automate everything? Yes. The parameters appear in your DAW's automation list, including the MORPH X / Y.

Glossary

- **Ghost.** One FM operator (an oscillator with its own envelope). There are six.
- **Haunt.** One ghost frequency-modulating another. Also the name of the first macro (FM growl).
- **NOISE ghost.** A noise source that can be heard or can haunt the ghosts.
- **ADD SAMPLE / SAMPLE ghost.** The sample slot. The sample can haunt a ghost, be heard, or be a ghost's waveform.
- **Macro.** One of the six big knobs: HAUNT, SPECTRE, POSSESS, DRIFT, VEIL, CURSE.
- **SUMMON.** Makes a sound from your words.
- **Drift.** The main slow LFO of the synth, also a macro.
- **Wobble.** A drawn, tempo-synced shape that moves a target.
- **Squeeze.** A three-band squash for a loud, bright finish.
- **Comb.** A tuned, metallic resonance.
- **Snapshot.** A saved copy of the whole sound in the SNAPSHOTS bank.
- **Board / puck / orbit.** The MORPH tab: a board of sounds, the puck that blends them, and the automatic movement of the puck.
- **Low Mono.** Makes everything below a chosen frequency mono.
- **Clean Sub.** A sub that skips filters and effects.